Audit Ledger — Session 17

Date: Session 17 (continuation of Session 16)
Scope: All files with status VERIFIED, PORTED, or PARTIAL in MANIFEST.csv
Audit method: Java source method-count comparison, golden test pass/fail analysis, logic review of targeted implementations

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Summary

Category	Files	Defects found	Defects fixed	Tests added
VERIFIED	~29	0	0	+9 new groups
PORTED	~24	0	0	0
PARTIAL	10	0	0	0
Total	**~63**	**0**	**0**	**+9**

New groups verified this session: 9
BLOCKED → COVERED moves: 9 (plane., mth.mulAndTruncate, mth.rayIntersectsAABB, mth.rotationAroundAxis)
Test count:* 180 → 189 (debug + release)

PARTIAL Files

`net/minecraft/core/Direction.rs` — PARTIAL

Java: 119 methods (overloaded counting), **Rust:** ~80+ methods
Gap analysis:
- `Direction.Plane` — was a `todo!()` stub. **IMPLEMENTED** this session.
- Added `Plane` enum (Horizontal=0, Vertical=1)
- Added `PLANE_VALUES`, `HORIZONTAL_FACES`, `VERTICAL_FACES`, `HORIZONTAL_AXES`, `VERTICAL_AXES`
- Methods: `ordinal`, `faces`, `axes`, `length`, `get_random_direction`, `get_random_axis`, `test`, `iterate`, `shuffled_copy`
- Added `Axis::get_plane()` method
- **Remaining gaps:** Codec fields (`CODEC`, `VERTICAL_CODEC`, stream codecs) — intentionally deferred per batch rules; not on parity-tested path.
- **Defects found:** None in existing code.
- **Golden result:** 6 new groups pass (plane.constants, plane.iterate, plane.test, plane.getRandomDirection, plane.getRandomAxis, plane.shuffledCopy)

`net/minecraft/core/BlockPos.rs` — PARTIAL

Java: ~90 public methods, **Rust:** ~77
Gap analysis:
- `clampLocationWithin` — in BLOCKED_ON_UNPORTED_TYPES; a constant discrepancy in the golden data makes it unresolvable (see the detailed note in BLOCKED_ON_UNPORTED_TYPES in parity_batch2.rs)
- Most other method gaps are private helpers or codec-related
- **Defects found:** None in existing code (all 7 iterator groups and all other groups pass)

`net/minecraft/core/Vec3i.rs` — PARTIAL

Java: ~43 public items, **Rust:** ~29
Gap analysis: Codec methods and DFU-related items intentionally not ported
Defects found: None (Vec3i groups all pass in parity_core.rs)

`net/minecraft/core/Rotations.rs` — PARTIAL

Java: 4 public methods, **Rust:** 5
Defects found: None

`net/minecraft/world/level/block/Rotation.rs` — PARTIAL

Java: ~11 methods, **Rust:** ~10
Gap analysis: `getRotated(int)` not ported (no parity test covers it)
Defects found: None

`net/minecraft/world/level/levelgen/MarsagliaPolarGaussian.rs` — PARTIAL

Java: 3 public methods (constructor, reset, nextGaussian), **Rust:** 3
Note: Previous session count of "7J vs 3R" was an overcounting error. All 3 real public methods are ported.
Defects found: None

`net/minecraft/world/phys/AABB.rs` — PARTIAL

Java: ~55 methods, **Rust:** ~55
Defects found: None (all 34 non-blocked AABB groups pass; `aabb.toString` stays BLOCKED on FloatingDecimal subnormal formatting)

`net/minecraft/world/phys/Vec2.rs` — PARTIAL

Java: 21 methods, **Rust:** 13
Gap analysis: Some constants and derived methods not ported; all parity-tested groups pass
Defects found: None

`net/minecraft/world/phys/Vec3.rs` — PARTIAL

Java: ~65 methods, **Rust:** ~52
Gap analysis: Some utility/codec methods not ported; `lerp(double, Vec3, Vec3)` still BLOCKED
Defects found: None (all vec3.* groups pass)

`net/minecraft/advancements/AdvancementType.rs` — PARTIAL

Java: 5 public methods, **Rust:** 4
Gap analysis: `getFrame()` (codec-adjacent) not ported
Defects found: None

VERIFIED Files

All 29 VERIFIED files were confirmed passing via golden tests before this session began (180 tests, all green in both debug and release mode).

`net/minecraft/util/Mth.rs` — VERIFIED

Gap analysis (pre-session 17): 3 methods previously BLOCKED:
- `mulAndTruncate(Fraction, int)` — BLOCKED on `Fraction` type
- `rayIntersectsAABB(Vec3, Vec3, AABB)` — BLOCKED on `Vec3`, `AABB`
- `rotationAroundAxis(Vector3fc, Quaternionf, Quaternionf)` — BLOCKED on JOML

Session 17 action: All 3 implemented. The blocking types were already ported:
- `Fraction` → `javacompat::fraction::Fraction`
- `Vec3`, `AABB` → PARTIAL but implemented
- JOML → `javacompat::joml::rotation_around_axis` (already existed)

Golden result: All 3 new groups pass. Header comment updated.

Remaining gaps:
- `getSeed(Vec3i)` — BLOCKED on Vec3i (partial)
- `lerp(double, Vec3, Vec3)` — BLOCKED on Vec3 (partial)

Other VERIFIED files

All other 28 VERIFIED files (RandomSource, random implementations, RNG files, ChunkPos, SectionPos, small enums, etc.) — no defects found via golden test analysis. All groups continue to pass.

PORTED Files

All 24 PORTED files checked via method-count comparison.
Key files:
- `Identifier.rs` — 47J vs 37R (gap: some static builders not ported, no parity tests cover them)
- `ARGB.rs` — 52J vs 50R (2 methods not ported, neither covered by parity tests)
- Marker interfaces / enums (ClickAction, LightLayer, RenderShape, etc.) — 0 Java methods, correct stubs

Defects found: None.

Fixes Applied This Session

1. **Direction.rs** — Implemented `Direction.Plane` enum (Horizontal, Vertical) with all 8 methods
2. **Direction.rs** — Added `Axis::get_plane()` method
3. **Mth.rs** — Implemented `mul_and_truncate(Fraction, i32) -> i32`
4. **Mth.rs** — Implemented `ray_intersects_aabb(Vec3, Vec3, AABB) -> bool`
5. **Mth.rs** — Implemented `rotation_around_axis(Vector3f, Quaternionf, Quaternionf) -> Quaternionf`
6. **parity_batch2.rs** — Added 9 new test functions (6 plane., 3 mth.)

7. **parity_batch2.rs** — Moved 9 entries from BLOCKED_ON_UNPORTED_TYPES to COVERED

Test Results

Mode	Passed	Failed	Ignored
Debug	189	0	1
Release	189	0	1

The 1 ignored test is `blockpos_clamp_location_within` (golden data contradicts itself — see note in BLOCKED_ON_UNPORTED_TYPES).

Known Remaining Gaps

File	Gap	Reason
`Mth.rs`	`getSeed(Vec3i)`, `lerp(double,Vec3,Vec3)`	Blocked on Vec3i/Vec3 partial status
`Direction.rs`	Codec fields	Intentionally deferred (DFU rule)
`BlockPos.rs`	`clampLocationWithin`	Golden data self-contradict-ory
`BlockPos.rs`	Some iterator methods	Not on parity-tested path yet
`AABB.rs`	`toString`	PORT-BLOCKED: FloatingDecimal subnormal formatting